

Working Groups (all)

GY457 Autumn Term 2025 (Seminar Topic 5)

Lecturer: *Dr Sara Bagagli*

Context¹

This seminar builds on the lecture on Topic 5: “Monocentric City Model II”. During the lecture, we have introduced the concept of bid-rents and discussed the comparative statics properties of the monocentric city model. We discussed how property prices adjust to compensate for changes in access to the CBD. The purpose of this seminar is to deepen our understanding of the model and explore its applicability to investment decisions. The idea is to illustrate how theory and evidence help developing expectations regarding property price effects of anticipated events with spatially differential effects, allowing for more targeted investments.

“Homework” to be carried out prior to the seminar

The compulsory readings for this seminar are:

Ahlfeldt, G. M. (2013): If We Build it, Will They Pay? Predicting Property Price Effects of Transport Innovations. *Environment and Planning A*. Vol 45(8).

Dubé, J., Thériault, M. & Des Rosiers, F., (2013): Commuter rail accessibility and house values: The case of the Montreal South Shore, Canada, 1992–2009. *Transportation Research Part A: Policy and Practice*, 54, pp.49–66.

Gibbons, S., & Machin, S. (2005). Valuing rail access using transport innovations. *Journal of Urban Economics*, 57(1), 148-169.

Optional reading

Gupta, A., Van Nieuwerburgh, S. & Kontokosta, C. (2022). Take the Q train: Value capture of public infrastructure projects. *Journal of Urban Economics*, 129.

All students are expected to read the above-mentioned papers, at least for the key ideas, intuition of the methodology and findings. All students must be prepared to answer the questions during the seminar. It is recommended that they prepare some notes. However, no submission of written answers is expected. Presenters, in addition, are expected to prepare slides for a joint presentation of their answers (one presentation per seminar). In doing so, they may split the tasks or work together on all tasks as they prefer.

¹The seminar was originally designed by Prof. Ahlfeldt and is kindly used with permission.

Scenario

During the seminar, the presenters will act as fund managers. Their task is to develop an investment fund focused on properties in areas where transit improvements, such as new underground stations or light rail expansions, are anticipated. The goal is to convince potential investors (the audience) that their fund represents an attractive investment opportunity. The fund should be “spatially blind,” meaning it should include a diverse portfolio of properties representative of the local market.

Seminar task:

1. Layout the theoretical rationale for why to expect property price effects due transport innovations; which economic model can be used to analyse how property prices vary in space; and how can it be adopted to predict proximity effects of a metro station.
2. Review the academic evidence on: how much do you expect property prices to increase (in percentage terms) in close proximity to a station following the opening of a new metro rail station (**discuss in detail how you arrived to your number**). Be aware that this is an investment pitch so think about how to incorporate the literature review appropriately.
3. Present evidence for your local market. Compare how a spatially targeted portfolio of properties near to-be-opened metro stations is likely to perform compared to an overall market portfolio. The investors will be interested in an annualised expected return and risk. The idea is to take the expected return (rent uplift) from the literature and compute the risk from the data set on Moodle.

Assuming that total return can be calculated by focusing on changes in asset value (the net operating income is small), use the dataset (GY457_Topic_5_Seminar_Data) available on Moodle to calculate the variables in the table on the next page and answer the following questions.¹

Does the risk in d) make intuitive sense? How can the result be rationalised using economic theory? To answer this question, consider the general impact of diversification on the risk of a portfolio and the extent to which the two competing portfolios are diversified.

4. Based on your answers to 1), 2) and 3), summarise why investors should invest in your product. Use your broader knowledge of urban economics to complement your argument.

¹The data in the file GY457_Topic_5_Seminar_Data.xlsx is an annual panel of market values of 1,000 properties in your local market between 1990 and 2017. Some of them are located close to a to-be-opened metro station (station_area = 1) while others are randomly located away from transportation hubs. The number of stations did not change over the period since the new stations are yet to be opened (at year $t > 2017$).

	Variable to be calculated:	Source:
a)	The average return on the market (the annual return should be based on the change in the value of a portfolio consisting of all properties in the market)?	Data on Moodle
b)	The average annualised effect of being invested in properties close to a new metro station during the opening period (the annualised “causal effect”)? Assume that to realise the full premium associated with a metro station opening, a property must be held for five years. Acknowledge the uncertainty that surrounds the metro rail capitalisation effect and provide a lower-bound, a central, and an upper-bound estimate.	Literature review from question 2.
c)	The average annualised return expected from investing into a property where a train station will be built.	Combine a) with b)
d)	The risk, expressed as the standard deviation of returns based on the change in the value of a portfolio over time for two portfolios: i. All properties in the market (the value of the portfolio is the sum of all its assets). ii. Only properties near a train station (the value of the portfolio is the sum of all assets near a train station).	Data on Moodle
e)	The estimated effect of focusing on listed properties close to to-be-opened stations on the risk-adjusted return as measured by the Sharpe ratio (for a brief introduction/definition, see this page)	Combine a) c) and d) with the risk free rate approximated by a 10 year bond .

During the seminar

The seminar will be as follows:

1. Discussion. After an introduction all students will vote on whether they think the investment idea is feasible before listening to the presenters’ pitch.

2. Presentation.

- a. The presentation may cover up to 60 minutes. Note: 1) The audience is encouraged to ask questions especially if the answer is unclear; 2) All presenters should carry an approximately similar weight in the presentation; 3) During the presentation, it is expected that the presenters “engage” with the audience, i.e., prompt the audience to suggest answers; and 4) The audience is expected to challenge the answers provided by the presenters; 5) Presenters

should discuss how they executed the numerical exercises in some detail and in a manner that is accessible to the audience.

- b. After the presentation, the investors (the audience) will cast a second vote to reveal whether or not they have been convinced by the investment strategy and if they would invest in the fund.

After the seminar

If indicated by the feedback during the seminar, presenters will revise their presentation. Within 24 hours, they will then ask the lecturer to upload the slides onto the GY457 Moodle page. Please, use the following filenames to save your presentations: Topic_1_group_X, where X indicates the seminar group.